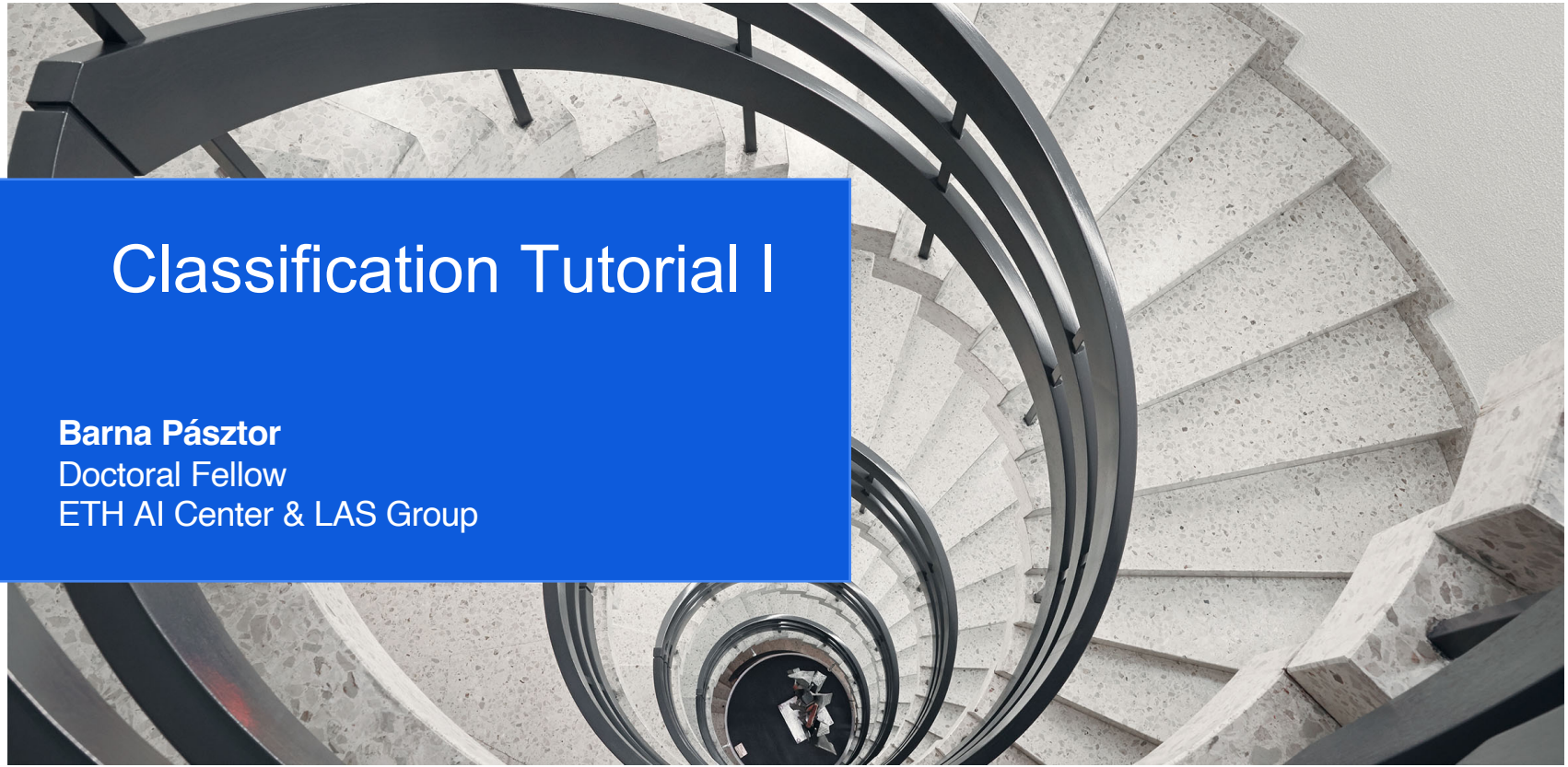
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Classification Tutorial I

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IML Tutorial: Classification

1. Hard-Margin SVM
2. Types of Errors in Classification
3. Decision Thresholds and the ROC curve
4. Fairness in Machine Learning

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1. **Hard-Margin SVM**
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Hard-Margin SVM: Problem Setup

Problem
Find a classifier that discriminates between 2 classes.

Data: $\mathcal{D} = \{x_i, y_i\}_{i=1}^{N}$ such that $x_i \in \mathbb{R}^n$ and $y_i \in \{-1, +1\}$

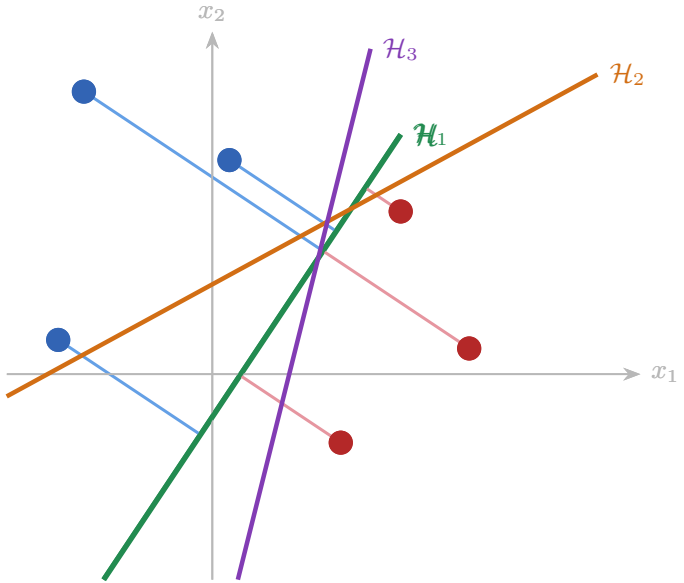
Example: Cats vs. Dogs

- $x_i \in \mathbb{R}^n \rightarrow$ pixel intensities
- $y_i = +1 \rightarrow$ cat images
- $y_i = -1 \rightarrow$ dog images

Geometrical Intuition: Maximum Margin

Problem: Multiple lines can separate the two classes. Which one to choose?
Strategy: Find a separating hyperplane H with maximum margin
Key ideas:

- **Separating Hyperplane:** H puts all +1 points on one side and -1 on the other
- **Margin:** distance from H to the nearest data point of either class
- **Maximum Margin:** choose H to maximize the margin

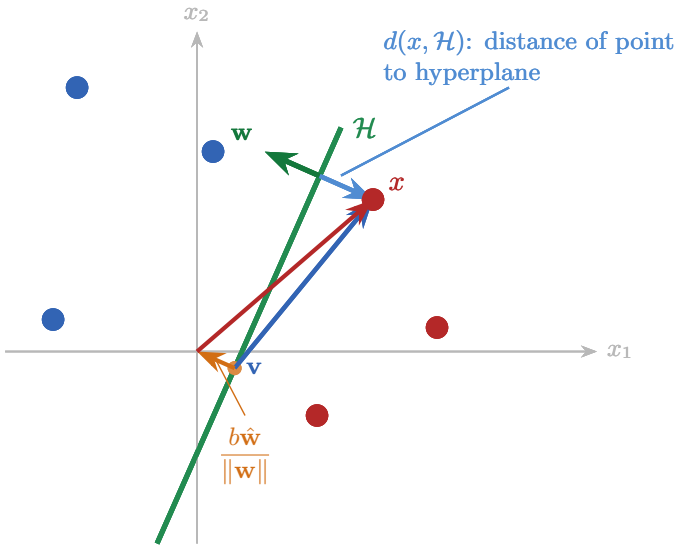


Mathematical Formulation – Distance to the hyperplane

Hyperplane: $\mathcal{H}_{w,b} = \{x | w^T x + b = 0, w \in \mathbb{R}^n, b \in \mathbb{R}\}$

Invariant to scale:

$\forall \alpha > 0, \alpha w$ and αb define the same hyperplane. Let $\hat{\mathbf{w}}$ denote the unit direction vector of $\mathbf{w}$.

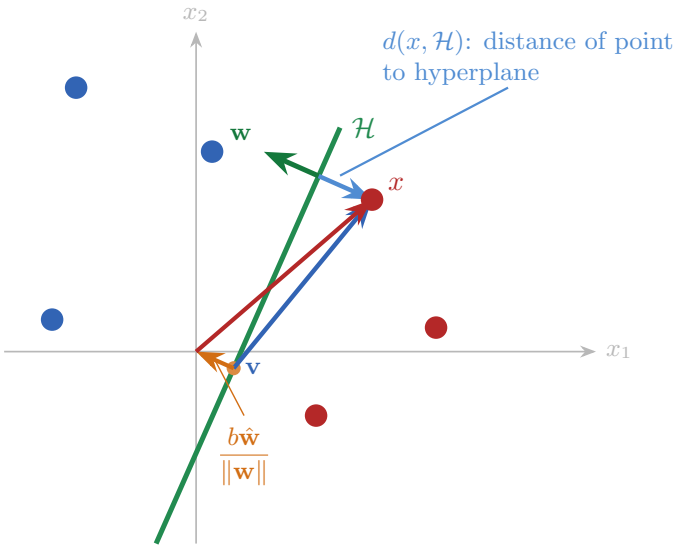


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Invariant to scale: $\forall \alpha > 0, \alpha w$ and αb define the same hyperplane

$$
\begin{aligned}
d(x, \mathcal{H}) &= v^T \hat{\mathbf{w}} * \text{sign} \\
&= \left(\frac{b\hat{\mathbf{w}}}{\|\mathbf{w}\|} + x \right)^T \hat{\mathbf{w}} * \text{sign} \\
&= \frac{b}{\|\mathbf{w}\|} + x^T \hat{\mathbf{w}} * \text{sign} \\
&= \frac{b + x^T \mathbf{w}}{\|\mathbf{w}\|} * \text{sign} \\
&= \frac{(b + x^T \mathbf{w}) y}{\|\mathbf{w}\|}
\end{aligned}
$$



Formulation of the Loss: SVM Objective

Max-margin objective: $\max_{\mathbf{w},b} \min_{1 \le i \le N} d(\mathbf{x}_i, y_i, \mathbf{w}, b) \qquad (\text{s.t. } \|\mathbf{w}\| = 1)$

Derivation (without loss of generality):

- **Step 1: Alternative formulation:** $\max_{\alpha} \alpha \text{ s.t. } d(\mathbf{x}_i, y_i, \mathbf{w}, b) \ge \alpha \quad \forall 1 \le i \le N$
- **Step 2: Substitute d formula:** $\max_{\alpha} \alpha \text{ s.t. } \frac{(b + \mathbf{x}_i^T \mathbf{w}) y_i}{\|\mathbf{w}\|} \ge \alpha \quad \forall 1 \le i \le N$
- **Step 3: Normalize by setting:** $\|\mathbf{w}\| = \frac{1}{\alpha}$

$$\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|} \text{ s.t. } (b + \mathbf{x}_i^T \mathbf{w}) y_i \ge 1 \quad \forall 1 \le i \le N$$

SVM Objective (Hard-Margin): $\min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 \text{ s.t. } (b + \mathbf{x}_i^T \mathbf{w}) y_i \ge 1 \quad \forall 1 \le i \le N$

Solving with Lagrange Multipliers I

Lagrangian:

$$\max_{\alpha} \min_{\mathbf{w},b} L(\mathbf{w}, b, \alpha) \text{ s.t. } \alpha_i \geq 0$$

$$L(\mathbf{w}, b, \alpha) = \frac{1}{2}\|\mathbf{w}\|^2 - \sum_{i=1}^{N} \alpha_i [y_i(\mathbf{w}^T\mathbf{x} + b) - 1]$$

Optimal solution:

Support Vectors

$$\mathbf{w}^* = \sum_{i=1}^{N} \alpha_i x_i = \sum_{i \in \mathcal{S}} y_i \alpha_i x_i \text{ where } \mathcal{S} = \{1 \leq i \leq N : \alpha_i \neq 0\}$$

$$L(\mathbf{w}, b, \alpha) = -\frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} y_i y_j \alpha_i \alpha_j \langle x_i, x_j \rangle - b \sum_{i=1}^{N} \alpha_i y_i + \sum_{i=1}^{N} \alpha_i$$

Solving with Lagrange Multipliers II

Decision function: $f(\mathbf{x}) = b\langle \mathbf{w}^*, \mathbf{x} \rangle$

$$= b + \sum_{i \in \mathcal{S}} \alpha_i y_i \langle \mathbf{x}, \mathbf{x}_i \rangle$$

- Decision boundary: $f(\mathbf{x}) = 0$

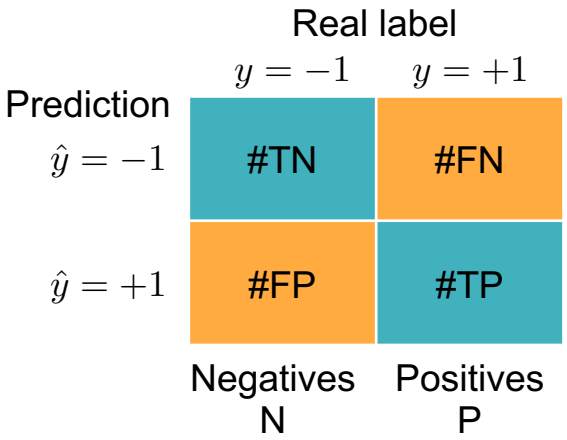
Key observation:

α and $f(x)$ depend on $\{x_i\}$ ONLY through dot-products $\langle x_i, x_j \rangle \implies$ kernelization! To be covered, next week.

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Recap: Types of Errors

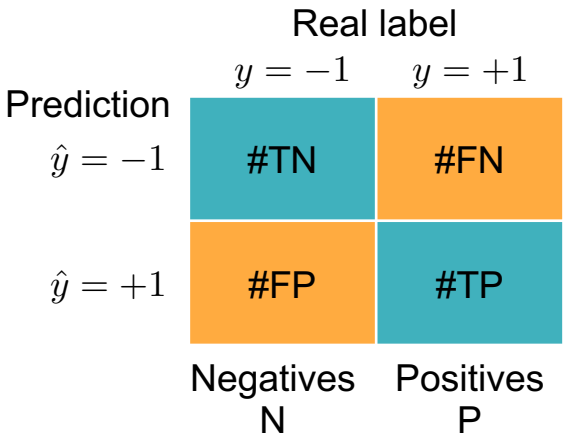


Assume y=1 means sick and y=-1 healthy.

- True positive rate (TPR): $\frac{\#TP}{\#P}$
- What percentages of sick people did we detect?
- False negative rate (FNR): $\frac{\#FN}{\#P} = 1 - \text{TPR}$
- What percentages of sick people did we not detect?

True positives, false negatives are both positives! => Divide by P

Recap: Types of Errors

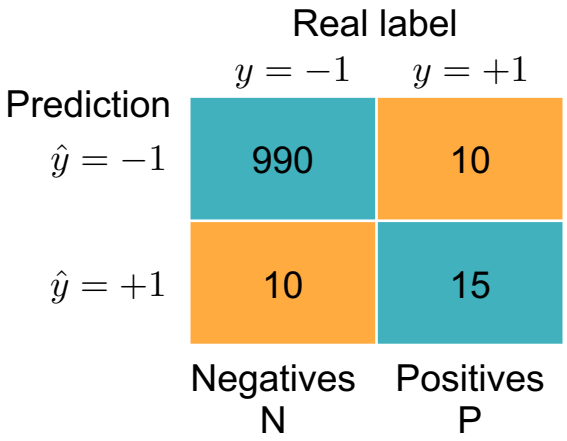


Assume y=1 means sick and y=-1 healthy.

- True negative rate (TNR): $\frac{\#TN}{\#N}$
- What percentages of healthy people did we identify?
- False positive rate (FPR): $\frac{\#FP}{\#N} = 1 - \text{TNR}$
- What percentages of healthy people did we misclassify as sick?

True negatives, false positives are both negatives! => Divide by N

Example: Covid Testing



Assume y=1 means sick and y=-1 healthy.

- FNR = $\frac{\#FN}{\#P} = \frac{10}{25} =$ 0.4
- FPR = $\frac{\#FP}{\#N} = \frac{10}{1000} =$ 0.01

- Same number of false positives and false negatives!
- But FNR and FPR are quite different -> many infection goes undetected!

What do we Call a “Positive”?

- Ultimately a convention

- Testing: “Unusual” event is positive
  - Covid Test: Patient has Covid
  - Fraud Detection: Credit card Transaction fraudulent
  - Spam Detection: Email is spam
  - Recidivism prediction: Criminal will re-offend
  - Fault detection: Product is defective
  - Predicting start up success: Startup is successful

What do we Call a “Positive”?

- “Z” - Detection/Test/Prediction:
  - Z is called positive

- Confusion: Detection often focuses on “bad” (i.e. negative) outcomes

The Cost of Different Errors

- Accuracy / 0-1 Loss treats all errors the same

- In which of these scenarios would you prefer more false positives over more false negatives?



  - Spam detection
  - Cancer screening
  - Predicting startup success
  - Predicting if defendant is guilty

Example: Testing for a rare disease

- We often care most about false negatives (did not detect cancer)
  - But false positives (wrong cancer diagnosis) are also bad!

		Real label $y=-1$	Real label $y=+1$
Prediction			
$\hat{y}=-1$		999,000	1
$\hat{y}=+1$		1000	9
		Negatives N	Positives P

Assume y=1 means sick and y=-1 healthy.

- FNR = $\frac{\#FN}{\#P} = \frac{1}{10} = 0.1$
- FPR = $\frac{\#FP}{\#N} = \frac{1000}{1000000} = 0.001$

- FPR is much smaller than FNR!
  - But almost all errors we make are false positives!

- Finding a good tradeoff is hard!
  - Ideally: Adjust it on the fly

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Adjusting the decision threshold

- Retaining model $\hat{f}$ for better trade-off between errors mightbe hard
  - Temporary data sccess
  - Large models are expensive to train
  - …
- Simple solution: Adjust the decision threshold
  - Default decision

$$\hat{y} = \begin{cases} +1 & \text{if } \hat{f} \geq 0 \\ -1 & \text{if } \hat{f} < 0 \end{cases}$$

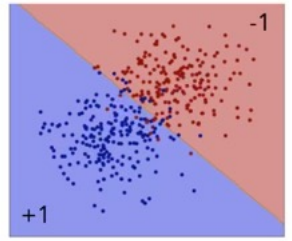
  - Threshold decison

$$\hat{y}_\tau = \begin{cases} +1 & \text{if } \hat{f} \geq \tau \\ -1 & \text{if } \hat{f} < \tau \end{cases}$$

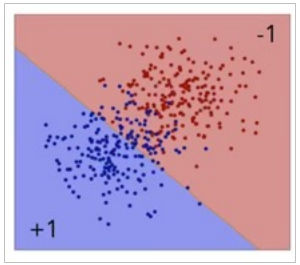
Adjusting the decision threshold

- Decision rule $\hat{y}_\tau = \begin{cases} +1 & \text{if } \hat{f} \geq \tau \\ -1 & \text{if } \hat{f} < \tau \end{cases}$
- Larger threshold
  - Predict $\hat{y} = -1$ more often
  - Fewer FP, more FN
- Smaller threshold
  - Predict $\hat{y} = +1$ more often
  - Fewer FN, more FP

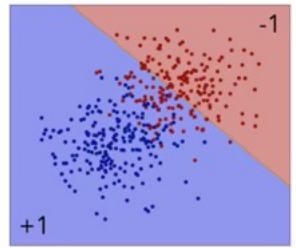
$\tau = 0$



$\tau > 0$

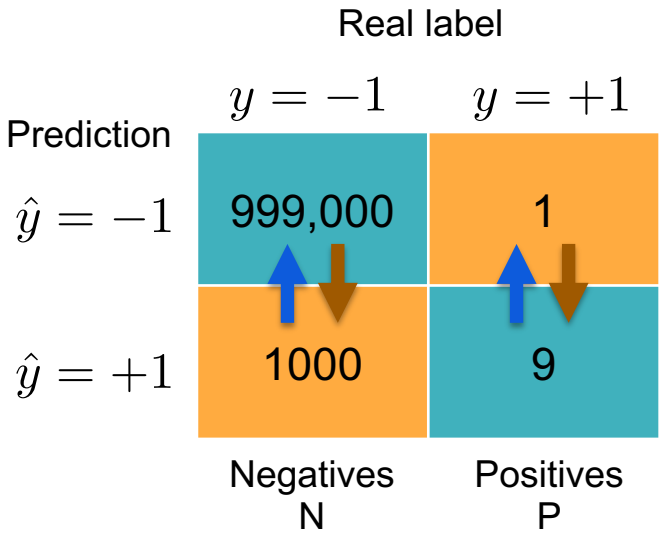


$\tau < 0$



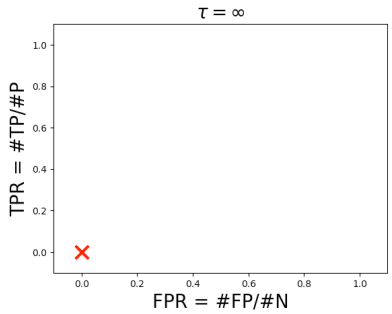
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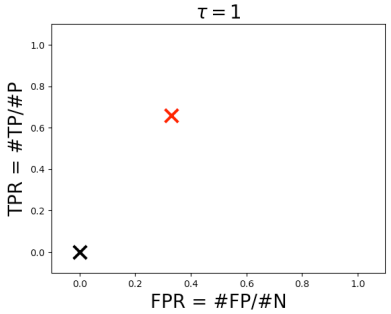


The ROC curve

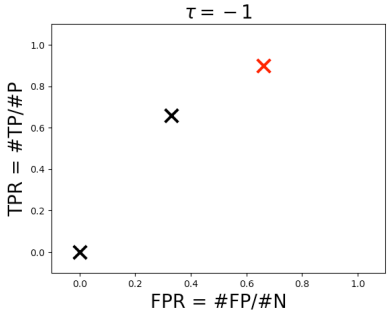
- For fixed $\hat{f}$, evaluate $\hat{y}_\tau$ for different τ.
  - Plot FPR($\hat{y}_\tau$) vs TPR($\hat{y}_\tau$) = 1 - FNR($\hat{y}_\tau$)



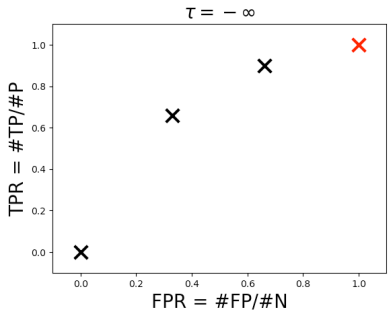
Large τ:
We never predict $\hat{y}_\tau = 1$!



Positive τ:
Few false positives
Not too many true positives



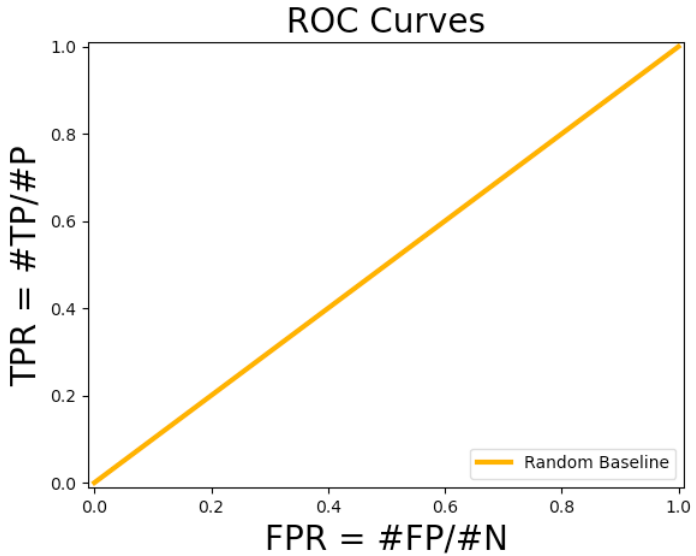
Negative τ:
More false positives
More true positives



Large negative τ:
We never predict $\hat{y}_\tau = -1$!

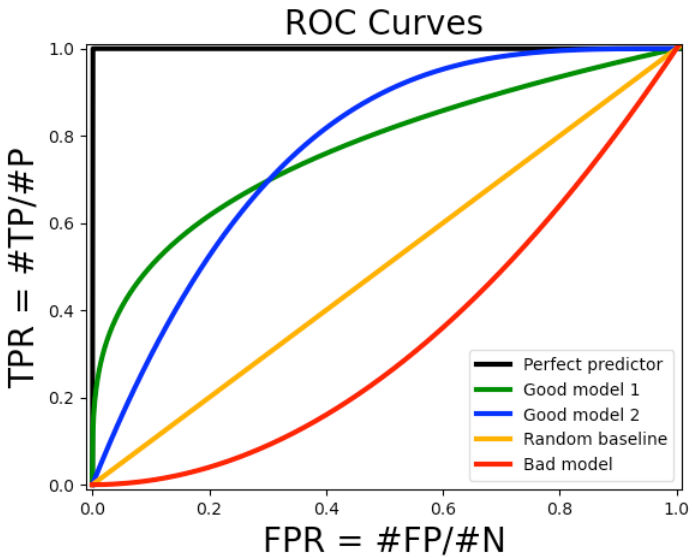
The ROC curve - Baseline

- Baseline: For each data point flip a (p-biased) coin
  - With probability p: $\hat{y}_\tau = 1$ (independent of real y)
  - p% of positives are classified as positive
    - => TPR = p
  - p% of negatives are classified as positive
    - => FPR = p
  - ROC curve should be above diagonal!



The ROC curve

- Different models might be better in different settings!
- One performance metric: AUROC
  - Area under the ROC curve
- Random model: AUROC = 0.5
- Perfect model ($\hat{y} = y$): AUROC = 1



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Fairness in Machine Learning

- Social context: Predictions about people
  - Actions based on predictions might discriminate
    - Example: Applicant pre-screening (Suitable for the job?)
      - Could be biased against applicants from certain countries

  - Formally: Sensitive attribute A, partitioning data points into groups G
    - Example: A=Nationality, G = {Swiss, Chinese, American, …}

Fairness in Machine Learning

- What kind of classifier would be “fair” in the applicant screening example?



A. Same FPR, FNR for all groups (equalised odds)
   - $\mathbb{P}(\hat{y}|Y = y, G = g)$ does not depend on g for fixed y

B. Only same FNR for all groups (equal opportunity)
   - $\mathbb{P}(\hat{y}|Y = 1, G = g)$

C. Same average prediction per group (demographic parity)
   - $\mathbb{P}(\hat{y}|G = g)$ does not depend on g

D. Individual predictions are invariant to group (individual fairness)
   - $\hat{f}(x, g)$ does not depend on g for fixed features x

Fairness in Machine Learning

- What kind of classifier would be “fair”?
    - Same FPR, FNR for all groups (equalised odds)
        - $\mathbb{P}(\hat{y}|Y = y, G = g)$ does not depend on g for fixed y
    - Same average prediction per group (demographic parity)
        - $\mathbb{P}(\hat{y}|G = g)$ does not depend on g

- There are many plausible definitions for fairness!

- Proposition [1]: If G is not independent of Y and both demographic parity and equalised odds hold, Y and $\hat{\mathrm{Y}}$ are independent.

[1]: Barocas, Solon, Moritz Hardt, and Arvind Narayanan.
Fairness and machine learning: Limitations and opportunities. MIT press, 2023.

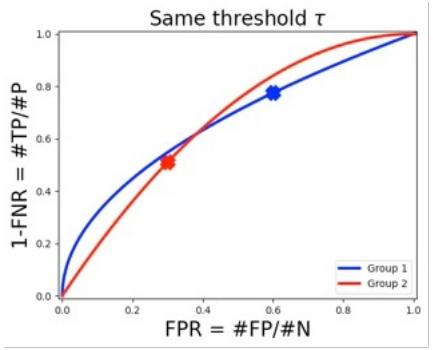
How to Achieve Fair Predictions

- Let’s say we decided to focus on equal opportunity (same FNRs for all g)
- How to achieve this?
  - Pre-Processing:
    - Modify training data
  - In-Processing:
    - Modify learning algorithm / Loss function
  - Post-Processing:
    - Modify how we extract predictions $\hat{y}$ from the model $\hat{f}$

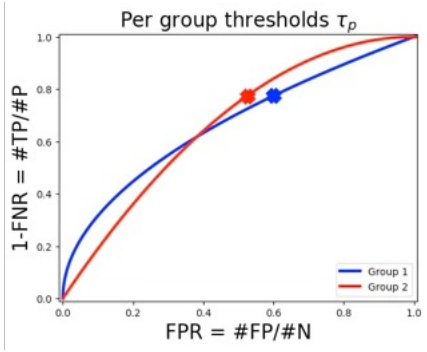
How to Achieve Fair Predictions

- Strong post-processing baseline: Group-wise thresholds
  - Assume we know group g for all inputs x
  - Choose different decision threshold τ_g for each group

$$\hat{y}(x, g) = \begin{cases} +1 & \text{if } \hat{f}(x, g) \geq \tau_g \\ -1 & \text{if } \hat{f}(x, g) < \tau_g \end{cases}$$



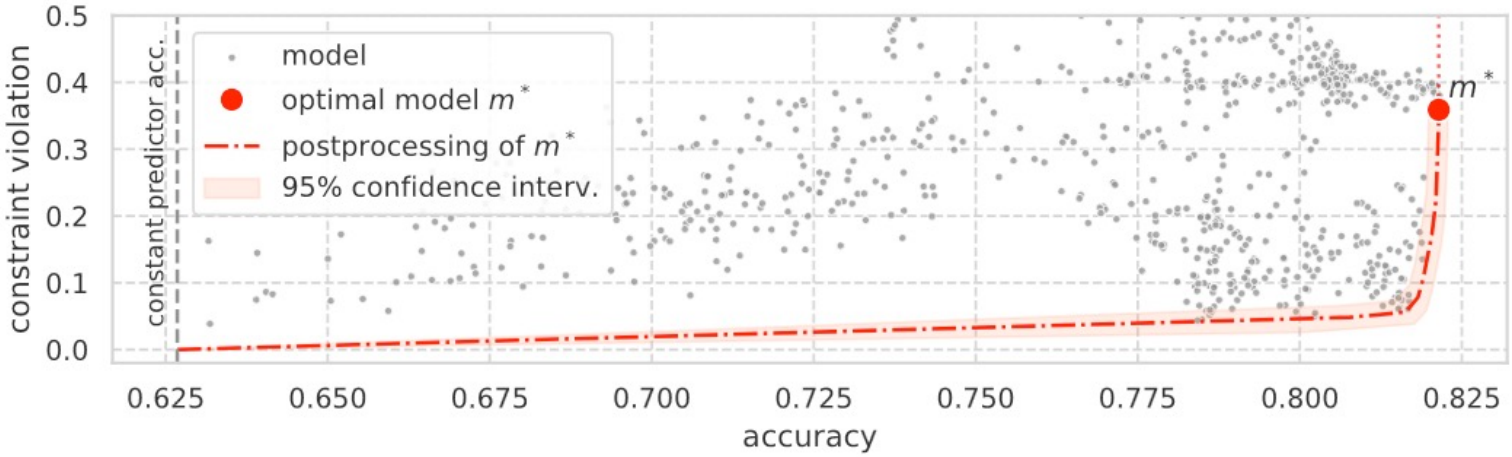
Same threshold: Different FNRs



Smaller threshold for group 1: Same FNR

How to Achieve Fair Predictions

- Post-processing reliably beats other methods on many datasets



Plot from:
Cruz, André, and Moritz Hardt. "Unprocessing Seven Years of Algorithmic Fairness."
The Twelfth International Conference on Learning Representations.